Geophys. J. Int. (1996) **126**, 513–524

A palaeomagnetic investigation of the Mashonaland dolerites, north-east Zimbabwe

M. P. Bates* and D. L. Jones

Department of Physics, University of Zimbabwe, PO Box MP167, *Mount Pleasant, Harare, Zimbabwe*

Accepted 1996 March 27. Received 1996 January 31; in original form 1995 August 11.

SUMMARY

Earlier palaeomagnetic work on the Mashonaland dolerite sills and dykes (1830 ± 230 Ma) showed that they record a geomagnetic field reversal. 28 new sites have been sampled in sills (26) and dykes (2) in north-east Zimbabwe. Detailed alternating-field demagnetization (AFD) is more effective in removing secondary components of remanence than thermal demagnetization (TD) and enables an assumed primary remanence to be isolated at 16 sill sites. Combined with earlier results there are 11 sill sites with a positive inclination (P) ($D=287°$, $I=53°$, $k=40$) and 18 with a negative inclination (N) ($D=114°$, $I=-47°$, $k=35$). The data lead to a positive class-B reversal test. Field tests suggest that the N remanence is the younger and that the period of Mashonaland igneous activity spanned a reversal of the Earth's magnetic field from P to N. The areal distribution of polarity is not random: sills with only P remanence are found in the more elevated central region of the Mashonaland Igneous Province, whilst predominantly N-bearing sills occupy the lower surrounding terrain.

Key words: Mashonaland dolerites, palaeomagnetism, Proterozoic, Zimbabwe.

1 INTRODUCTION

The Mashonaland sills (Fig. 1) are a suite of Early Proterozoic intrusive dolerites that crop out across much of the north-eastern quarter of the Zimbabwe Archaean craton. Although they mostly intrude basement granites (Fig. 1) they also intrude the Archaean greenstone belts. The youngest formation they intrude is the Great Dyke, which was emplaced at the Archaean–Proterozoic boundary. Their outcrop pattern appears to be dictated by topography and they, along with the Zimbabwe craton, have suffered little tectonic movement since intrusion. An outline of their petrology is given in McElhinny & Opdyke (1964).

In this paper, we present new palaeomagnetic data from the sills and associated dykes that show how igneous activity was temporally associated with tectonic activity in the nearby Limpopo Belt at approximately 1900 Ma. The palaeomagnetism of the Mashonaland Sills was first studied by McElhinny & Opdyke (1964), in which 13 of 15 sites sampled were found to carry one of two seemingly anti-polar remanence directions (omitting site 12, see later). The Mazowe River dyke (MD, Fig. 1) has a remanence direction (declination $D=98°$, inclination $I=-55°$, $N=5$, $R=4.99$, $k=403$, $\alpha_{95}=4°$; Wilson, Jones & Kramers 1987) very close to that of the Mashonaland Sills mean direction, while that of the Sebanga Poort dyke (SPD, Fig. 1) (Jones, Robertson & McFadden 1975) suggests that it also was emplaced temporally close to the Mashonaland Sills. The NNW-striking Sebanga dykes, which include the SPD, are believed to be feeders for the Mashonaland Sills (Wilson *et al.* 1987; Wilson 1991); the east-striking Mazowe dyke swarm, which includes the Mazowe River dyke, may have performed the same function. The combination of Mashonaland sills, Mazowe River and Sebanga dyke activity is called the Mashonaland Igneous Event (MIE; Wilson *et al.* 1987).

The best estimate of the age of the MIE is a Rb/Sr whole-rock date of 1830 ± 230 Ma from a Mashonaland sill (Compston & McElhinny 1975, recalculated by Wilson *et al.* 1987). In addition, there is a Rb/Sr date from the Mazowe River dyke of 1870 ± 600 Ma (Wilson *et al.* 1987). The assertion by McElhinny & Opdyke (1964) that the Mashonaland Sills post-date the approximately 1100 Ma Umkondo dolerites was based on a result from their site 12; the site was later re-visited and sampling was extended, and this led to a re-interpretation that removed the inconsistency in the relative ages of the two dolerite suites (Wilson *et al.* 1987). Site 12 has been omitted from the present analysis. The MIE is linked to contemporaneous activity in the Limpopo and Magondi Belts adjacent to the Zimbabwe craton. Limpopo Belt Central Zone granulites acquired an MIE remanence as they cooled after uplift (Morgan & Briden 1981; Morgan 1985). NW-directed thrusting and dextral shearing along the northern margin of the Central Zone at approximately 2000 Ma (Campbell *et al.* 1991)

* Now at: Sander Geophysics Ltd, 93 Legget Drive, Kanata, Ontario K2K 2B1, Canada

© 1996 RAS

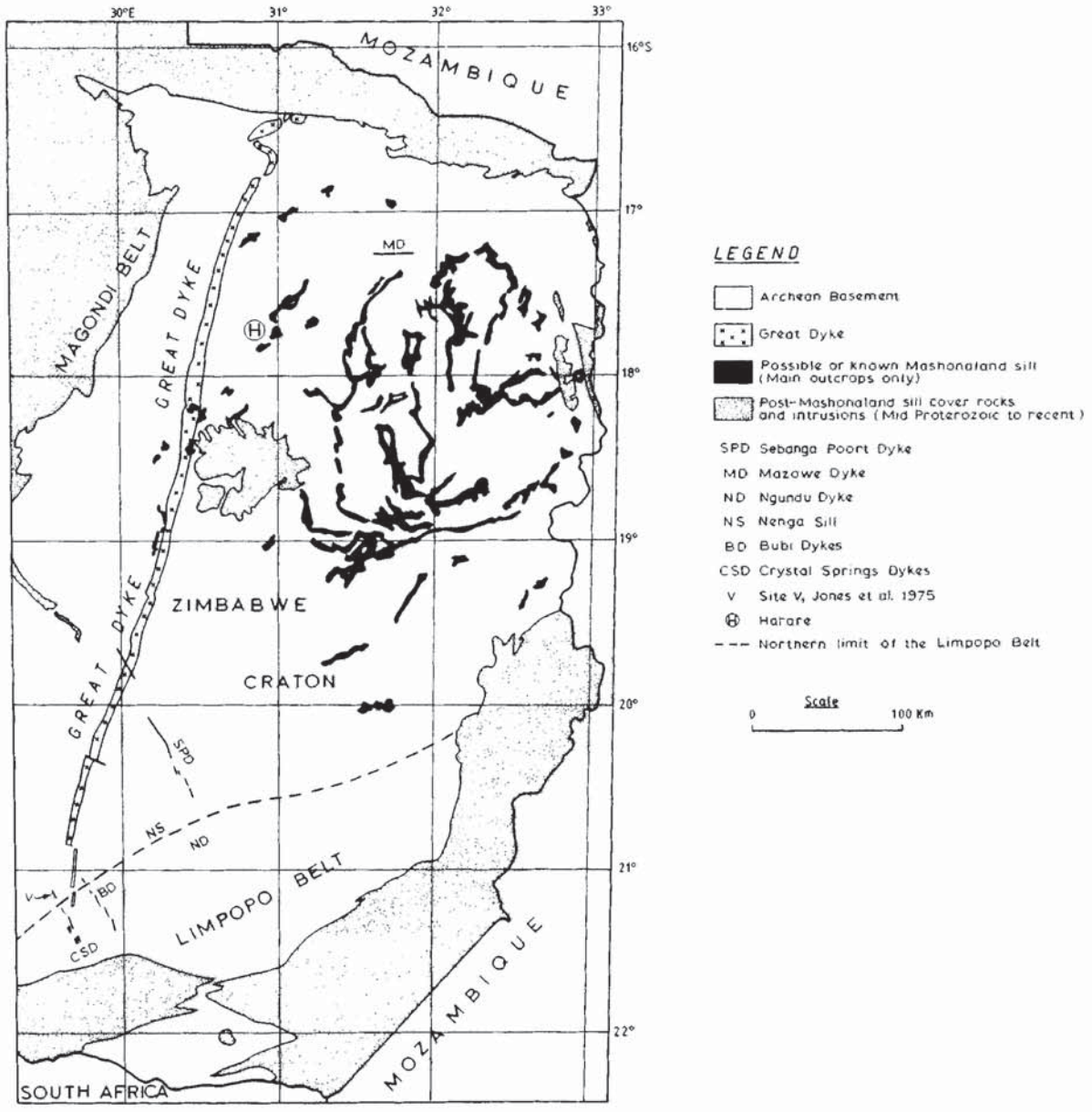


Figure 1. Summary map of the geology of eastern Zimbabwe (after Campbell *et al.* 1991 and the Zimbabwe Geological Survey provisional geological map 1991).

resulted in ENE–WSW-directed extension of pre-existing fractures in the Zimbabwe craton: this accommodated Sebanga dyke emplacement (Wilson 1991). The trend of the Sebanga dykes is also consistent with foreland extension related to SE-directed thrusting in the Magondi Belt at about 1800 Ma. Furthermore, local easterly thrusting in the northern part of the Magondi Belt is reflected in the easterly trend of the Mazowe dyke swarm (Campbell *et al.* 1991).

2 SAMPLING

Samples were collected from suspected Mashonaland sills at 26 new sites in north-east Zimbabwe (Fig. 2, Table 1). A dyke (SD2) intersecting one of the sills was sampled at site MS15, and three parallel dykes were sampled at site SD1 (Fig. 2). Many sites are in shallow valleys or 'vleis', where the top of the sill appears to form the valley floor. Other sites are in river beds that were easily accessible due to the prevailing drought. Hill tops and prominent exposures were avoided to minimize the risk of remagnetization due to lightning strikes, of which there is a high incidence in the area. At least five samples were collected at each site using a petrol-powered hand drill, except at site SD1, which was block sampled and subsequently drilled. Samples were oriented in the field using a sun compass, or a magnetic compass when overcast (MS25 only). Each drill-core sample was cut into two or three 2.4 cm diameter, 2.5 cm long specimens, which were immersed in dilute hydrochloric acid to remove metal particles acquired during drilling and cutting.

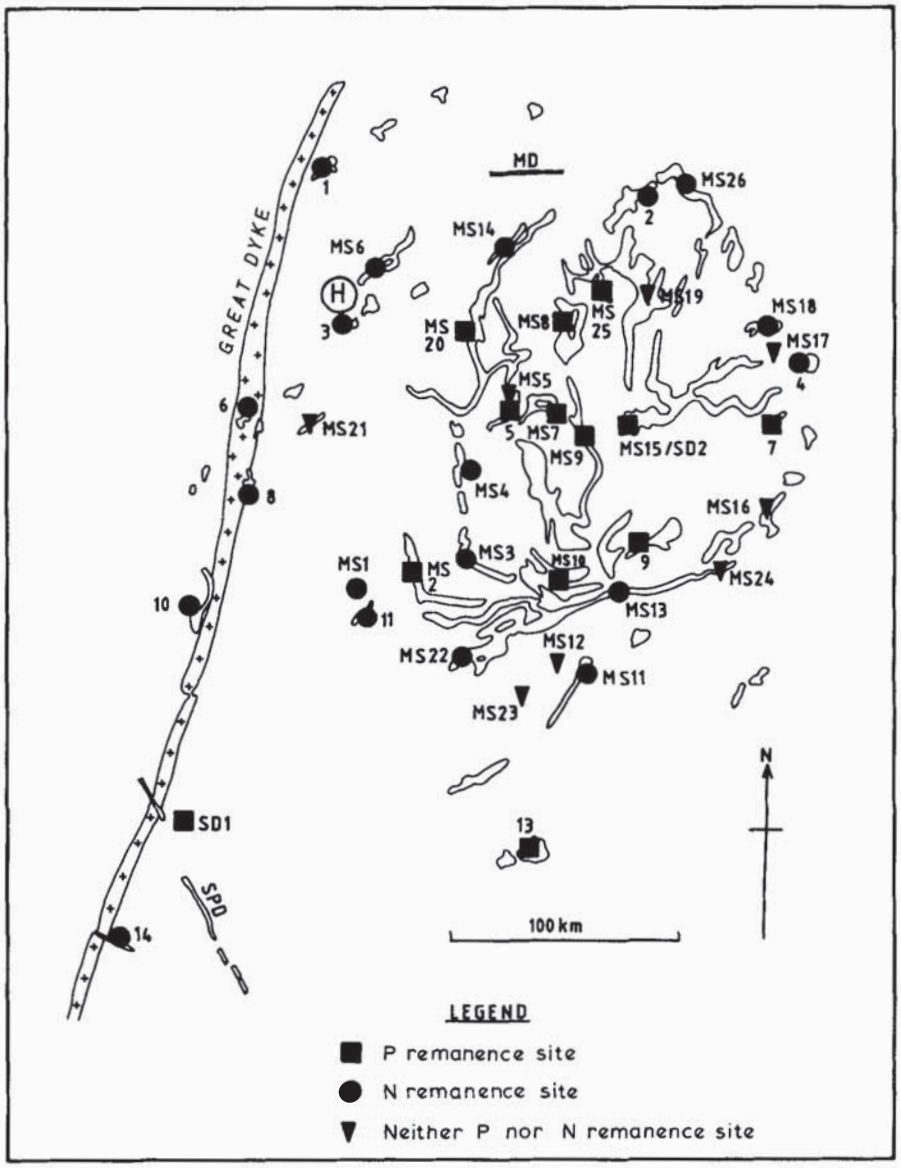


Figure 2. Location of palaeomagnetic sample sites from this study (prefixed MS or SD) and that of McElhinny & Opdyke 1964 (labelled with number only).

When combined with the studies of McElhinny & Opdyke (1964) and Wilson *et al.* (1987), the total sample set of 41 sites provides extensive coverage of the Mashonaland dolerite suite.

3 LABORATORY TECHNIQUES

A specimen from each core sample was subjected to detailed stepwise alternating-field (AF) demagnetization up to a maximum of 100 mT using a two-axis tumbling AF demagnetizer (McElhinny 1966). In addition, a specimen from each site was thermally demagnetized up to 700 °C using a Magnetic Measurements Thermal Demagnetizer. Thermally treated specimens were demagnetized first in AF at 10 mT in order to remove very high-intensity, low-coercivity components. Remanence directions were measured using a Molspin Minispin Magnetometer and a PAR SM-1 Spinner Magnetometer.

Linear vectors in the demagnetization data, which represent components of magnetization, were determined using LINEFIND (Kent, Briden & Mardia 1983). Where the data did not define a stable endpoint, CIRCLE DISPERSE (McFadden & McElhinny 1988) was used to analyse remagnetization circles (Jones *et al.* 1975), and this information was combined with linear vector data to calculate a site mean. Remagnetization circle data were not used if at least five linear vectors were available from separate core samples.

4 PALAEOMAGNETIC RESULTS FROM SILLS

Thermal demagnetization up to 500 °C has little effect on the remanence of the sills. Rapid decay of intensity occurs as

Table 1. Site locations and site mean directions of high coercivity remanence components from 26 Mashonaland Sills (MS) and 2 Sebanga dykes (SD).

Site	Lat/Long (°N/°E)	Dec (°)	Inc (°)	$N(n)$	R	k	α_{95} (°)	Plat (°N)	Plong (°E)	Pol
MS1	−18.90/30.90	116	−50	5(–)	4.86	29	15	11	339	N
MS2*	−18.87/31.18	296	62	4(4)	3.95	19	41	5	350	P
MS3	−18.85/31.38	111	−35	5(4)	4.96	61	12	12	327	N
MS4*	−18.42/31.38	124	−44	1(–)	–	–	–	–	–	N
MS5	−18.18/31.65	40	−75	5(3)	4.98	114	8	−38	9	–
MS6	−17.60/31.03	115	−34	4(2)	3.92	26	21	17	327	N
MS7	−18.17/31.76	270	43	5(2)	4.84	18	20	−8	326	P
MS8	−17.87/31.80	299	52	5(–)	4.96	90	8	13	343	P
MS9	−18.23/31.86	287	39	5(–)	4.98	246	5	8	328	P
MS10	−18.88/31.82	285	60	5(–)	4.99	591	3	−2	345	P
MS11	−19.27/31.86	104	−66	5(5)	4.93	22	24	−5	351	N
MS12	−19.23/31.80	–	–	–	–	–	–	–	–	–
MS13	−18.96/31.97	151	−69	5(3)	4.97	92	9	14	14	N
MS14	−17.42/31.60	94	−35	5(4)	4.98	90	10	−2	321	N
MS15	−18.28/32.17	296	59	7(–)	6.94	100	6	7	348	P
MS16	−18.55/32.75	201	−65	4(4)	3.88	8	67	22	48	–
MS17	−17.87/32.68	356	−50	6(–)	5.98	277	4	−77	48	–
MS18	−17.80/32.58	129	−30	4(4)	3.94	16	46	29	334	N
MS19	−17.73/32.17	167	−26	5(4)	4.91	23	20	56	9	–
MS20	−17.90/31.43	293	47	5(–)	4.93	60	10	11	336	P
MS21	−18.26/30.77	–	–	–	–	–	–	–	–	–
MS22	−19.23/31.37	105	−48	5(2)	4.94	48	12	3	334	N
MS23	−19.35/31.59	–	–	–	–	–	–	–	–	–
MS24	−18.91/32.38	–	–	–	–	–	–	–	–	–
MS25	−17.63/31.95	292	43	3(2)	3.00	369	11	11	333	P
MS26	−17.25/32.38	119	−43	5(–)	4.89	38	13	17	336	N
SD1	−19.85/30.13	318	7	5(–)	4.85	26	15	43	325	P
		140	−53	1(–)	–	–	–	–	–	N
SD2	−18.28/32.17	125	−73	13(–)	12.39	20	10	1	7	N

Notes: $N(n)$: total number of specimens included in the mean (of which n only provide remagnetization circle data, the rest provide vector data); R: length of resultant of N unit vectors; k: best estimate of precision parameter; α_{95}: semi-angle of cone of 95 per cent confidence; Plat and Plong: palaeopole latitude and longitude; Pol: polarity N (negative) or P (positive) of MIE remanence. *: sites not included in the mean of Mashonaland Sills due to lack of vector data or only one specimen in the mean.

580 °C is approached, indicating that magnetite is the principal magnetic mineral (Fig. 3). Some specimens retain a small but unstable remanence between 580 and 680 °C indicating the presence of minor haematite. The narrow unblocking-temperature (T_{ub}) range of the natural remanence in the sills means that it is difficult to resolve components of magnetization using this technique. Therefore, AF data have been used exclusively to derive site means.

AF demagnetization typically removes high-intensity (sometimes greater than 100 A m^{-1}), low-coercivity (0–50 mT) remanence components of variable direction (Fig. 4). At a few sites the low-coercivity remanences are consistent within site, but there is no correlation from site to site or with any known remanence direction for this region. The random direction, high T_{ub}, high intensity characteristics of the low-coercivity remanence indicate that it is probably an isothermal remanent magnetization (IRM) resulting from lightning strikes (Graham 1961). Removal of the low-coercivity remanence generally isolated one of two approximately anti-polar high-coercivity remanence directions (P and N: positive and negative inclination, respectively) at 18 sill sites (Fig. 5, Table 1). Two of these sites (MS2 and MS4) have been omitted from the mean of sill sites, either because only one specimen provided reasonable data, or because only remagnetization circle data were available. Data from this study and that of McElhinny & Opdyke (1964) are combined when referring to mean P and mean N directions in the text (Fig. 5, Table 2).

Four sites (MS12, MS21, MS23 and MS24) exhibited no stable high-coercivity component. Four more sites had high-coercivity components other than P or N. MS17 could carry a recent viscous remanence (present Earth's field has $D=353°$, $I=-53°$) or a Triassic-age Karoo-related remanence. Since there are no known Karoo-age rocks in the area, and the site was highly weathered, the former interpretation is preferred. MS19 may be an Umkondo-age remanence (McElhinny & Opdyke 1964); again, a remagnetization origin is likely, since the site is part of an extensive sill within the area dominated by MIE intrusions, and site MS25 from the same sill complex has a P remanence. Our sites MS5 and MS16 could be N-bearing sites with unusual declinations, but MS5 comes from a sill where McElhinny & Opdyke (1964) found a P remanence (their site 5) that falls well within the grouping of other P sites. The remanence at site MS16 is similar to that in the Great Dyke but these bodies are physically far apart and a correlation is unlikely. The origins of the remanences at MS5 and MS16 remain uncertain.

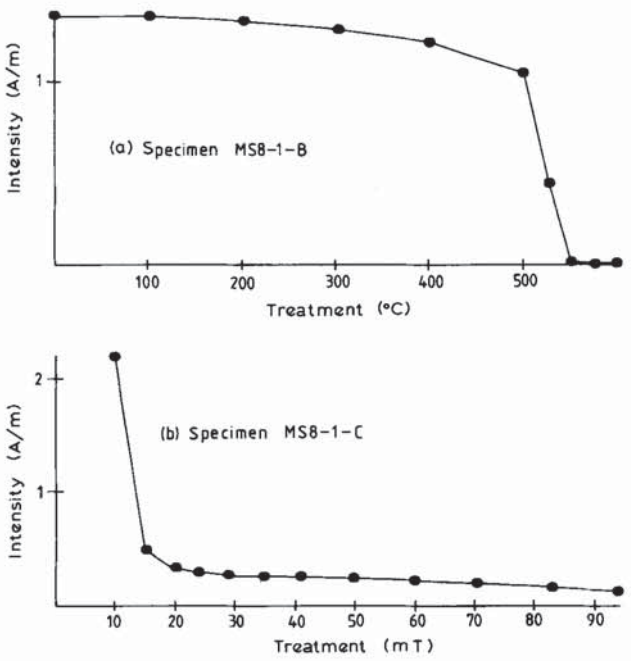


Figure 3. Examples of response of magnetic intensity to (a) thermal and (b) AF demagnetization for the Mashonaland sills. Both specimens were cut from a single core. (a) shows thermal demagnetization characteristics after initial AF demagnetization at 10 mT.

5 PRIMARY ORIGIN OF P AND N

P and N remanence directions have been determined from dolerites believed to be either Mashonaland sills or related Sebanga dykes from sites across the entire eastern half of the Zimbabwe craton. Several lines of evidence favour the interpretation that P and N are primary magnetizations acquired at or soon after emplacement of the sills, as opposed to later secondary remagnetizations due to a regional metamorphic event, slow cooling due to uplift or delayed cooling at depth (Zhang & Halls 1995). These are as follows:

(1) P and N have not been identified together at any given site;

(2) N-bearing sills intrude the 2461 ± 16 Ma Great Dyke of Zimbabwe, which has a distinctly different primary magnetization ($D = 220°$, $I = -58°$, $N = 7$, $R = 6.89$, $k = 56$, $\alpha_{95} = 8°$; Jones *et al.* 1975);

(3) average within-site dispersion ($k = 153$ and 87 for P and N) is lower than between-site dispersion ($k = 40$ and 35 for P and N), indicating that secular variation was averaged between sites, rather than within sites, and that each site effectively represents a spot reading of the palaeofield rather than a time-averaged reading (Morgan 1976);

(4) although there is a regional pattern in the polarity of the sills' remanence (see below), sills of opposite polarity are closely associated and do not occur exclusively at particular topographic levels. In addition to this evidence, data have been collected from two dykes that support a primary origin for P and N.

6 PALAEOMAGNETIC RESULTS FROM DYKES

Site SD1

This site, exposed in the dry bed of the Chuni River, comprises an approximately 28 m wide basaltic dyke (SD1A) with a strike of 340°, flanked to the west by 10 cm and 40 cm wide parallel basaltic dykes (SD1B and SD1C), intruding a granitic country rock (Fig. 6). Thermal and AF demagnetization of SD1A and SD1C removed poorly resolved low-T_{ub}, low-coercivity remanence components of unknown origin, and isolated high-T_{ub}, high-coercivity remanence components, which have been combined into a single mean and interpreted as a shallow P direction (Table 1) similar to the remanence at two sites from the SPD (Mushayandebvu 1991). This same remanence is also present in the country rocks adjacent to the dykes. Granite specimens 2.3 m west of SD1C (equivalent to 9.1 m west of SD1A) or greater had a variety of unrelated remanence directions. The data suggest that the P remanences in dykes SD1A and SD1C are related to the dykes themselves, because the country rock appears to have been remagnetized next to the dykes. The data do not comprise a full baked-contact test because there is no consistent remanence in the country rocks away from the dykes, and no zone of overlapping country rock/dyke remanence has been identified in the contact area. Nevertheless, the P remanence is likely to be primary

Table 2. Mean palaeomagnetic directions and palaeopoles of the Mashonaland Sills.

Polarity	Dec (°)	Inc (°)	N	R	k	α_{95} (°)	Plat (°N)	Plong (°E)	R	K	A_{95} (°)
P(1)	288	49	7	6.90	61	8	6	337	6.88	50	9
N(1)	114	−46	9	8.64	22	11	11	338	8.57	19	12
P & N(1)	292	48	16	15.53	32	7	9	338	15.44	27	7
P(2)	284	59	4	3.89	26	18	−3	345	3.83	18	22
N(2)	113	−48	9	8.88	67	6	10	337	8.85	54	7
P & N(2)	291	52	13	12.71	41	7	6	339	12.59	30	8
P(3)	287	53	11	10.75	40	7	3	340	10.66	30	9
N(3)	114	−47	18	17.52	35	6	11	337	17.43	30	6
P & N(3)	292	49	29	28.22	36	5	8	338	28.02	29	5

Notes: (1) this study; (2) McElhinny & Opdyke (1964); (3) a combination of both. NB McElhinny & Opdyke's results have been recalculated omitting their site 12 (see text). Abbreviations as for Table 1, except K and A_{95} are used instead of k and α_{95} when averaging palaeopoles.

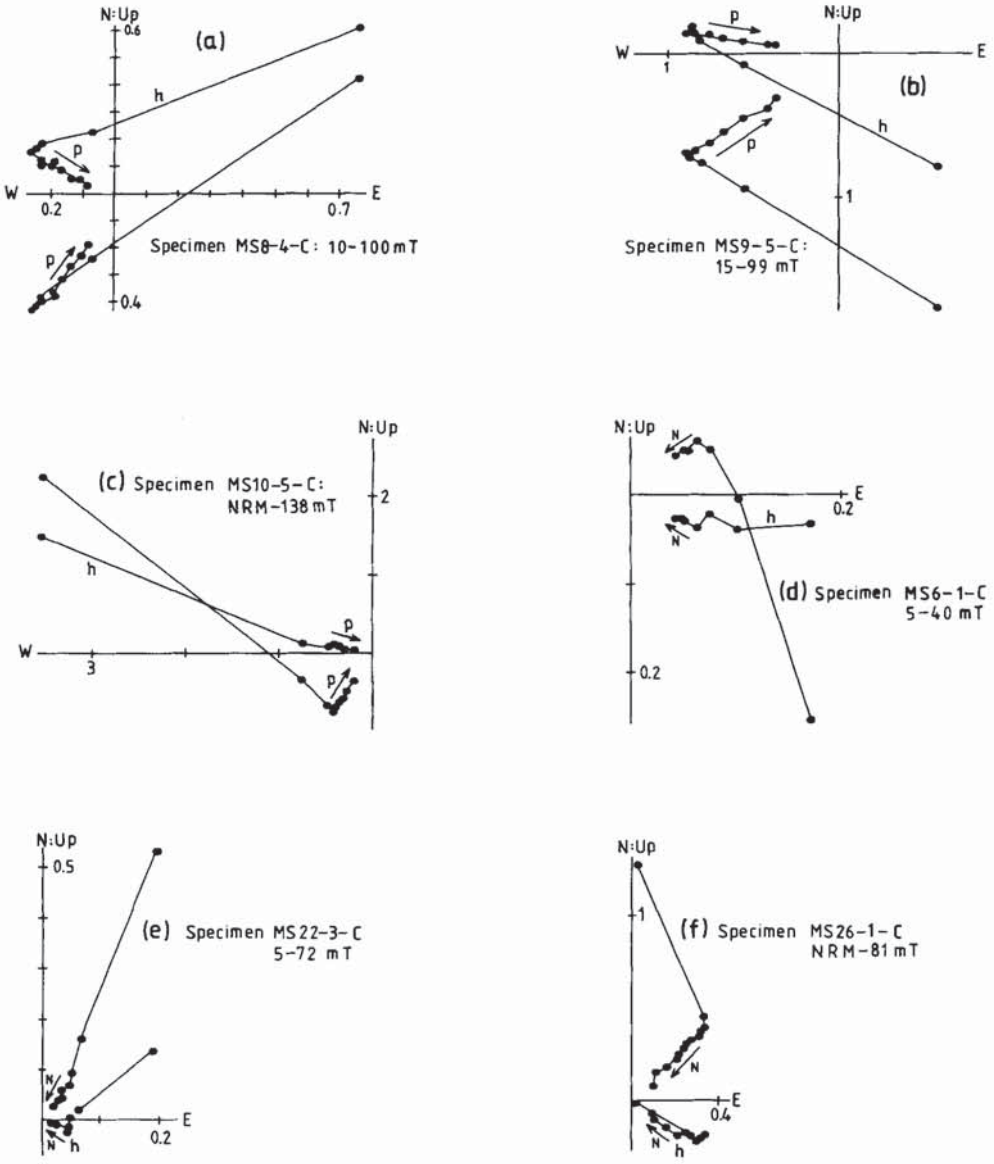


Figure 4. Examples of AF demagnetization data from the Mashonaland sills. Arrows are in the direction of increasing AF treatment and indicate stable endpoint data for components P (a–c) and N (d–f). Units for axes of the orthogonal vector plots are A m^{-1}. Projections onto the horizontal plane are denoted by h. The range of treatment displayed is given in mT for each specimen.

(i.e. acquired at or soon after the formation of the dykes during cooling) and the adjacent country rocks have either a thermal remanent magnetization (TRM) or a chemical remanent magnetization (CRM) due to heating and/or metamorphic fluids related to the injection of the dykes.

In addition, SD1B, which lies between the other two dykes and within the zone of remagnetization of the country rocks, carries a single-component, highly stable N remanence, unaffected by SD1A, which is only 2.7 m away. The inference is that this dyklet was emplaced after intrusion of SD1A and SD1C. The remanence direction of the dykes at site SD1 suggests that they are Sebanga dykes, and that dyke intrusion spanned at least one reversal of the Earth's magnetic field in which the sense was from P to N in terms of the local palaeofield.

Site MS15/SD2

This site comprises a 1.4 m wide basaltic dyke (SD2), striking 035°, within a dolerite sill (MS15, Fig. 7). The remanence of this site is complex, and includes the following features:

(1) a well-resolved high-coercivity, high-T_{ub}, P component from sill specimens away from the dyke;

(2) an N component from seven of the 10 dyke specimens, which also occurs in sill specimens adjacent to the dyke, with lower T_{ub}s and coercivities than P where they occur in the same specimen;

(3) a component X ($D=293°$, $I=22°$, $N=6$, $R=5.79$, $k=24$, $\alpha_{95}=14°$) with a shallower inclination than P, determined from two dyke specimens and four sill specimens adjacent to

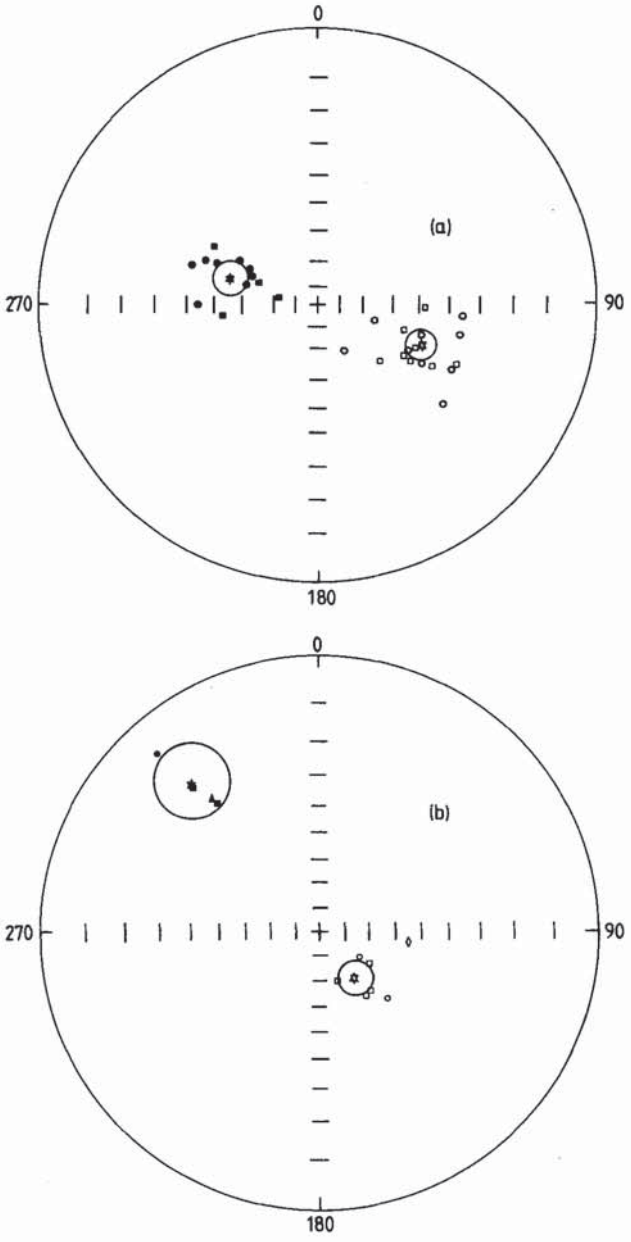


Figure 5. Equal-angle stereographic plot of site-mean P and N directions from (a) Mashonaland sills (circles: this study; squares: McElhinny & Opdyke 1964) and (b) Sebanga dykes (circles: this study; squares: Jones *et al.* 1975; triangles: Mushayandebvu 1991) and the Mazowe dyke (diamond: Wilson *et al.* 1987). Means of site means are shown by stars with circles of 95 per cent confidence. Closed (open) symbols denote positive (negative) inclinations.

the dyke, with lower T_{ub}s than N where they occur together in both the dyke and the sill.

P and X have not been determined in the same specimen; this may be due in part to their similar directions. P is clearly related to the sill, and the N component is clearly related to the dyke (Table 1). The separation of P and X is justified on the basis of their different relationship to the N remanence and different directions at the 95 per cent confidence level (using the technique of McFadden & Lowes 1981). Our preferred interpretation, which accounts for the spatial distribution and relative magnetic properties of the three remanence directions, is that P represents a primary TRM acquired during cooling of the sill, followed by intrusion of the dyke (after a field reversal), which acquired a primary N TRM and remagnetized the sill close to the dyke. The sill within 1 m of the margin was completely remagnetized, but only low-T_{ub} magnetite 1.5 m from the dyke was affected, and N is absent in the sill beyond 2 m from the margin. At an unspecified later date, metamorphic fluids exploiting the weakness along the dyke margin (Halls 1991) resulted in a low-temperature thermochemical remanence X with a patchy distribution in and around the dyke.

Our observations lead us to conclude that P and N were acquired by sills and dykes as primary magnetizations at the time of the MIE. Some, perhaps all, P-bearing bodies predate N-bearing bodies.

7 REVERSAL TEST

A reversal test (McFadden & McElhinny 1990) has been carried out on the combined data set, in which we compare the population of P site means with the population of the antipodes of N site means. We find that it is not possible to reject the hypothesis that the two populations share a common precision parameter, and that there is sufficient data in each population (number of sites >4) to test the hypothesis that the data are drawn from a single population. The critical angle (gamma_c) between the populations at which a common mean can be rejected is 9.2°. The observed angular difference is approximately 8°, so that a common mean cannot be rejected. This constitutes a positive class B ($5° < \text{gamma}_c < 10°$) reversal test. Therefore, we can conclude that secular variation has been averaged out in the means of sites and that the Earth's magnetic field changed polarity during the time that P and N were acquired. Given the paucity of geomagnetic data for the Precambrian, this observation adds credence to the supposition that the Earth's magnetic field had assumed its axial dipolar geometry by the early Proterozoic (see also Halls 1991).

8 MAGNETIC ANISOTROPY

Measurements of anisotropy of magnetic susceptibility have identified an oblate magnetic fabric in the sills with vertical minimum susceptibility axes at most sites (Mushayandebvu, Bates & Jones 1995a). Although the degree of anisotropy in these rocks is mostly small (less than 10 per cent), this raises the possibility that remanent inclination may be shallower than the ambient field. Dispersion of site-mean data from sills appears approximately isotropic (Fig. 5), but least-squares planes fitted to both P and N populations are nearly vertical, indicating a greater variation in observed inclination than declination. These patterns are consistent with the effects of anisotropy due to the tabular shape of the sills as described by Abrahamsen (1986); magnetic vectors are deflected towards the plane of the sill but deflection varies as a function of the angle between the ambient magnetic field and the sill plane. This may account for at least some of the difference between the mean directions of the N-bearing sills and dykes (see below). Precise predictions of the amount of deflection, if any, are difficult to make without a good knowledge of conditions,

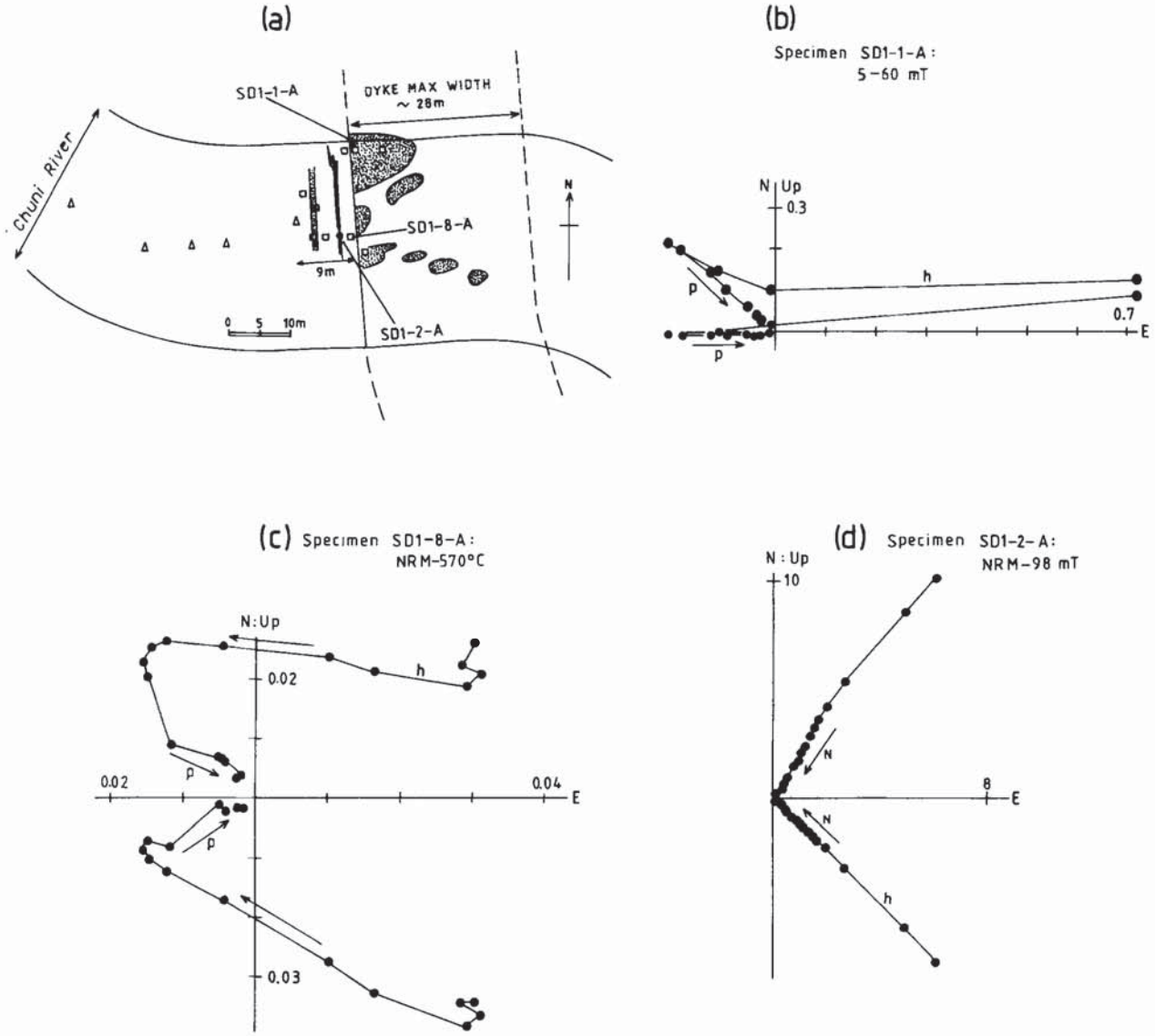


Figure 6. (a) Sketch map of dykes SD1 showing the sample locations. Closed circle: N remanence; open squares: shallow-P remanence; open triangles: unrelated remanences. (b)–(d) Examples of AF demagnetization data for samples from SD1 on orthogonal vector plots. Symbols and units as for Fig. 4.

for example magnetic susceptibility and geomagnetic field intensity, at the time of remanence acquisition.

9 DISCUSSION

The increased data base for the Mashonaland sills has confirmed the identity of many suspected Mashonaland sills plus two Sebanga dykes. It is apparent from Fig. 2 that the distribution of polarity in the sill population is not random. There is a distinct area in the centre of the sills that has exclusively P remanence, which more or less coincides with the highest-elevation exposures. The central P zone is surrounded by sites that are predominantly N, although a few P sites also occur. The pattern suggests that intrusion during P polarity times took place at generally higher structural levels (dyke SD2, however, indicates that some magma also reached high levels during N polarity times). Primary remanence does not survive metamorphism at upper greenschist levels or higher (Pullaiah *et al.* 1975). Assuming a present-day continental geotherm (if anything, the geotherm was steeper in the Proterozoic), the lower depth limit of sill emplacement is approximately 20 km.

Antipolarity of mean P and N data indicates that intrusion of sills occurred over a sufficient time period to average secular variation. Since remanence is probably primary, this means that intrusion of the sills took place over a considerable time span. There are no data recording the intermediate geomagnetic field during the reversal of polarity, suggesting that the intrusion of P- and N-bearing sills was separated in time. It is unlikely that the two polarities represent single sills; rather, they represent at least two separate phases of sill intrusion.

The remanence directions observed thus far in Sebanga dykes, presumed feeders to the sills, are similar to, but not identical to, those of the sills (Fig. 5). Six N-bearing Sebanga dyke sites have a mean direction ($D = 142.3°$, $I = -63.8°$, $R =$

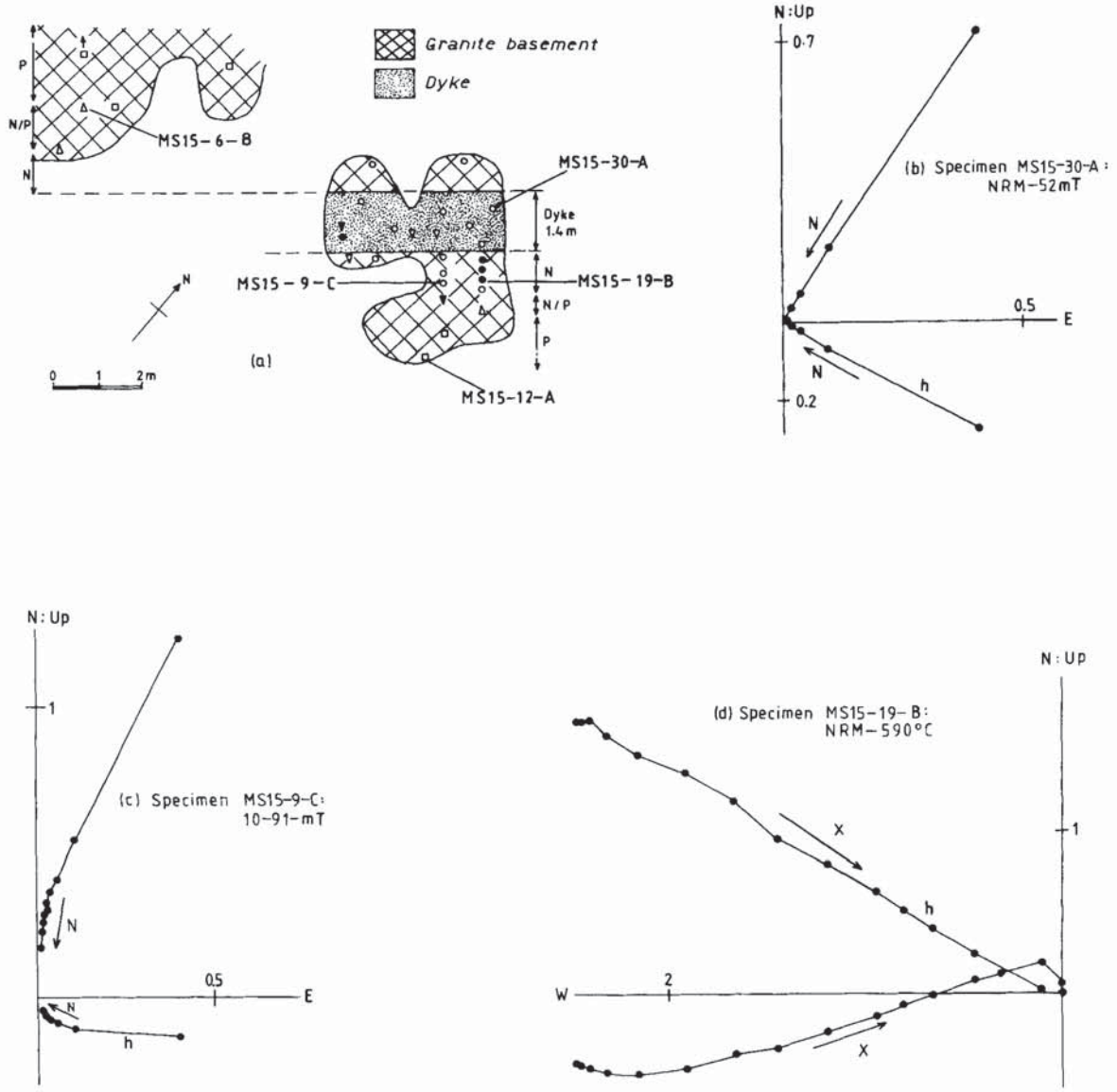


Figure 7. (a) Sketch map of dyke SD2 intruding sill MS15 showing the sample locations and the distribution of components N, P and X. Open circles: N remanence; open squares: P remanence; open triangles: P and N remanence; inverted open triangles: no P or N present; solid symbols indicate the presence of X remanence. (b)–(h) Examples of AF demagnetization data for samples from SD2/MS15 on orthogonal vector plots. Symbols and units as for Fig. 4.

5.93, $k=83$, $\alpha_{95}=7.9°$) significantly different from that of the 18 N-bearing sills (Table 2, Fig. 5). If site SO (Mushayandebvu 1991) is excluded (its direction is poorly defined) and site V of Jones *et al.* (1975) is included then four shallow-P-bearing dykes have a mean direction ($D=319.7°$, $I=19.3°$, $R=3.96$, $k=83$, $\alpha_{95}=10.1°$) clearly different to that of the 11 P-bearing sills (Table 2, Fig. 5) (but close to the X direction observed at site SD2). This does not mean that the Sebanga dykes are not feeders to the sills. Only a few dykes have been located and analysed and these are not the ones that supplied material to the sills. The only dyke sampled from the main area of sill exposure was at site MS15, and emplacement of this dyke post-dated the sill. The actual feeder dykes to the sills have yet to be located.

Uplift and cooling of the Limpopo Belt (LB) led to widespread remagnetization. Morgan (1985) identified an early phase of remagnetization at 1980 Ma (Limpopo metamorphics 'A') in the central and southern parts of the belt that resulted in a magnetization similar to, but distinct from, the Mashonaland sills. Later cooling in the northern part of the LB led to remagnetization of the southern end of the Great Dyke satellites (Jones *et al.* 1975) at 1895 Ma (Morgan, 1985) in a direction intermediate between the earlier remagnetization and the Mashonaland sills. The shallow P dykes from earlier studies are outside the LB but close to the boundary shear zone (Rollinson & Blenkinsop 1995) and may have been affected by a positive polarity remagnetization event in the northern LB, previously undetected due to a lack of data from northern LB gneisses. However, our site SD1 is well within the craton away from the LB, and indicates that the remanence

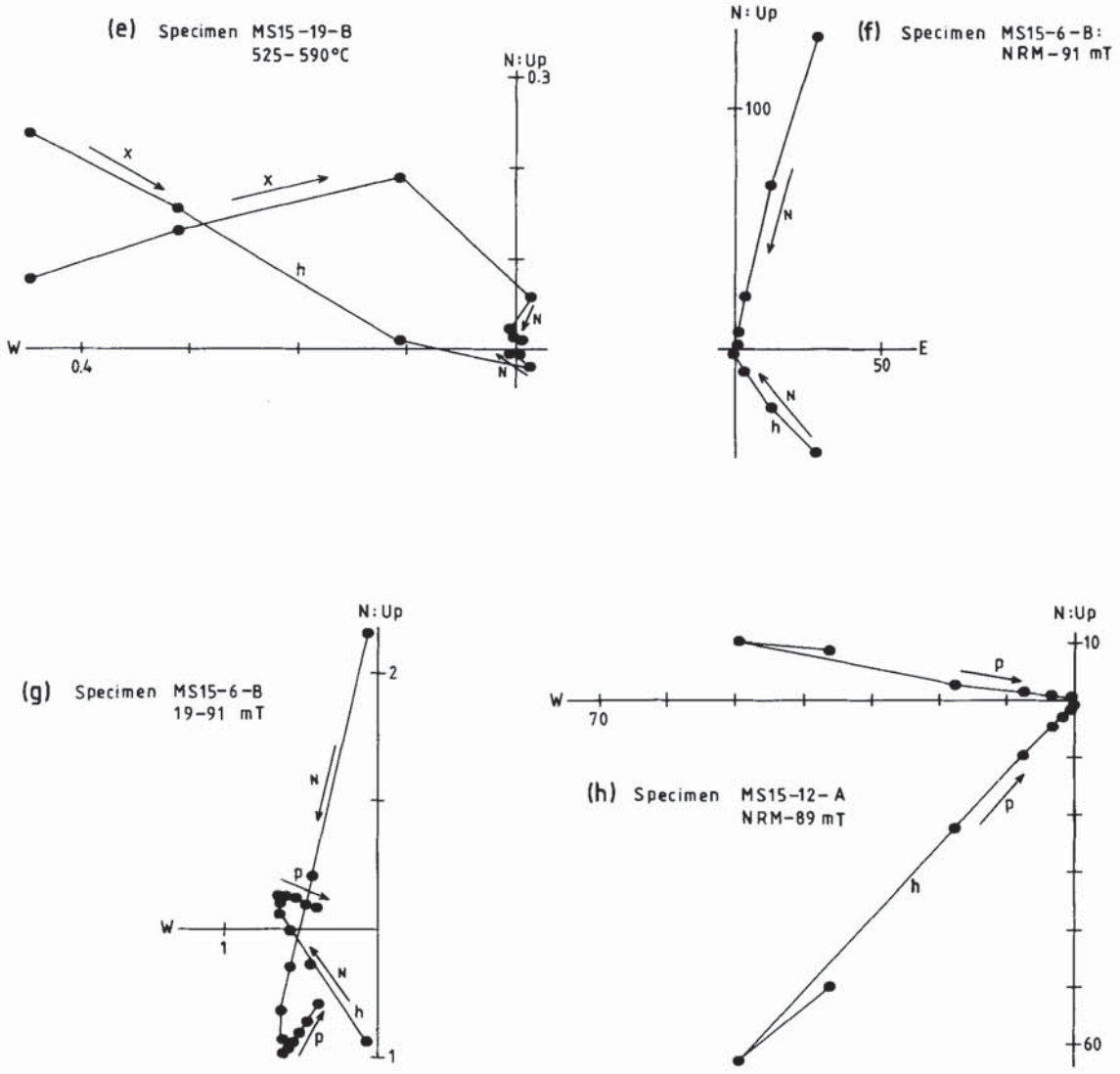


Figure 7. (*Continued.*)

in all these dykes is probably primary. This means that the P-bearing segment of the SPD was emplaced at a different time from the rest, but the actual time of emplacement is not well constrained. The only indication comes from site SD1, which suggests P Sebanga dykes predate the N Sebanga dykes.

Mushayandebvu, Jones & Briden (1995b) have drawn attention to the similarity of the mean palaeomagnetic directions of the Crystal Springs and Bubi dyke sets within the Northern Marginal Zone of the LB, the Nenga sill and Ngundu dyke (both are close to the northern boundary of the LB) and the N-bearing section of the SPD, and correlate the acquisition of their remanence. The remanence of the Crystal Springs and Bubi dykes is probably primary (Jones *et al.* 1975), but the Nenga and Ngundu bodies were probably emplaced in Great Dyke times and completely remagnetized at around 1900 Ma. Mushayandebvu *et al.* (1995b) give a mean direction for these five bodies (Group A) ($D=151.2°$, $I=-50.2°$, $R=4.94$, $k=70$, $\alpha_{95}=9.2°$) that is significantly different from that of the Mashonaland Sills. This suggests a different age of acquisition of remanence, but Mushayandebvu *et al.* (1995b) were unable to decide whether the bodies acquired their remanence before or after the sills.

The Zimbabwe and Kaapvaal cratons were closely associated during the Proterozoic (Layer 1986) and, with the LB, formed a single unit known as the Kalahari craton. Fig. 8 shows the apparent polar-wander path (APWP) for the latter for the period 2050–1800 Ma according to several authors (e.g. Jones *et al.* 1975, Morgan 1985, Hattingh 1986a,b,c). The path moves westwards at low positive latitudes from the time of emplacement of the Bushveld Complex until the intrusion of the Mashonaland sills. There is a large gap in reliable data until the intrusion of the Umkondo dolerites at 1080 Ma (Allsopp *et al.* 1989), so little is known about the APWP after the MIE. There is a great deal of data from the Kalahari craton for the period 2050–1800 Ma, but, because of uncertainties in pole positions and the ages of bodies (and acquisition of remanences), it is difficult to see anything other than the broad picture. If shallow-P-bearing Sebanga dykes generally predate

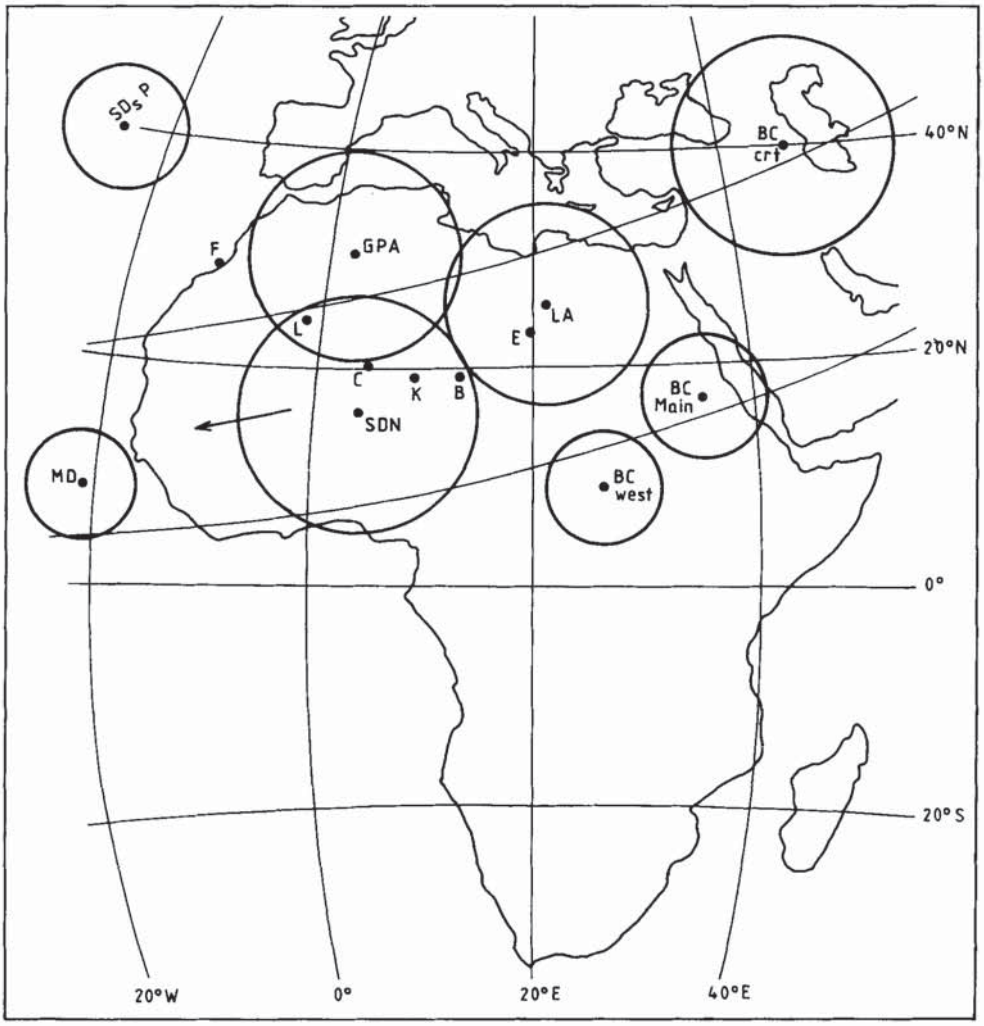


Figure 8. Apparent polar-wander path for the Kalahari craton for the period 2050–1800 Ma. MD: Mashonaland sills (this study); BC crt: Critical Zone, Bushveld Complex (Hattingh, 1986c); BC Main: Main Zone, Eastern Bushveld Complex (Hattingh 1986b); BC west: Main Zone, Western Bushveld Complex (Hattingh 1986a); LA: Limpopo metamorphics A (Morgan 1985); GPA: Group A (this study); SDN: Sebanga dykes (N bearing) (this study; Mushayandebvu 1991; Jones *et al.* 1975); SD_SP: Sebanga dykes (shallow P bearing) (as for (SDN); B, C, E, F, K and L: virtual geomagnetic poles from sites within the Limpopo Belt in satellite dykes of the Great Dyke (Jones *et al.* 1975).

the N-bearing Sebanga dykes and the remanences of the bodies of Group A predate the Mashonaland sills then the pole swung northwards to mid-latitudes and westwards before returning to low latitudes. If, however, the remanences in these bodies post-date that in the sills it is possible that the APWP does swing northwards after sill emplacement, as suggested by McElhinny *et al.* (1968).

10 CONCLUSIONS

The Mashonaland Sills have a primary palaeomagnetic remanence that occurs as one of two anti-polar directions at separate sites. Anti-polarity has been established via a class B reversal test. Sills emplaced at high structural levels have P remanence, whereas those emplaced at low structural levels have mostly, but not exclusively, N directions. Limited palaeomagnetic data from two Sebanga dykes suggest that P predates N, so that sill emplacement was at deeper structural levels with time. This may have come about due to changing levels of emplacement, or a constant level of emplacement during a period of uplift. N-bearing Sebanga dykes have an apparently similar but in fact significantly different remanence from N-bearing sills, with a similar situation for their P-bearing counterparts. Different sections of the Sebanga Poort dyke were emplaced at different times.

ACKNOWLEDGMENTS

The authors gratefully acknowledge NORAD and the Research Board of the University of Zimbabwe for research funding. Dr T. Blenkinsop (University of Zimbabwe) is thanked for helpful comments. J. E. C. Bates, T. Kapondo and S. Takaidza are thanked for assistance in the laboratory and the field. We also thank N. Abrahamsen and H. C. Halls, whose comments during the refereeing process improved the presentation of this report.

REFERENCES

Abrahamsen, N., 1986. On shape anisotropy, in *Twenty-five years of geology in Aarhus,* pp. 11–21, ed. J. T. Möller, *Geoskrifter,* **24,** Aarhus University, Aarhus, Denmark.

Allsopp, H.L., Kramers, J.D., Jones, D.L. & Erlank, A.J., 1989. The age of the Umkondo Group, eastern Zimbabwe, and implications for palaeomagnetic correlations. *S. Afr. J. Geol.,* **92,** 11–19.

Campbell, S.D.G., Oesterlen, P.M., Blenkinsop, T.G., Pitfield, P.E.J. & Munyanyiwa, H., 1991. A provisional, 1:250 000 scale tectonic map and the tectonic evolution of Zimbabwe. *Annal. Zimbabwe geol. Surv.,* **16,** 31–51.

Compston, W. & McElhinny, M.W., 1975. The Rb–Sr age of the Mashonaland dolerites of Rhodesia and its significance for palaeomagnetic correlation in southern Africa, *Precamb. Res.,* **2,** 305–315.

Graham, K.W.T., 1961. The remagnetization of a surface outcrop by lightning currents, *Geophys. J. R. astr. Soc.,* **6,** 85–102.

Halls, H.C., 1991. The Matachewan dyke swarm, Canada: an early Proterozoic magnetic field reversal, *Earth planet. Sci. Lett.,* **105,** 279–292.

Hattingh, P.J., 1986a. The palaeomagnetism of the Main Zone in the western Bushveld Complex, *Earth planet. Sci. Lett.,* **79,** 441–452.

Hattingh, P.J., 1986b. The palaeomagnetism of the Main Zone of the eastern Bushveld Complex, *Tectonophysics,* **124,** 271–295.

Hattingh, P.J., 1986c. The palaeomagnetism of the Merensky Reef footwall rocks of the Bushveld Complex, *Trans. Geol. Soc. S. Afr.,* **89,** 1–8.

Jones, D.L., Robertson, I.D.M. & McFadden, P.L., 1975. A palaeomagnetic study of Precambrian dyke swarms associated with the Great Dyke of Rhodesia, *Trans. Geol. Soc. S. Afr.,* **78,** 57–65.

Kent, J.T., Briden, J.C. & Mardia., K.V., 1983. Linear and planar structure in ordered multivariate data as applied to progressive demagnetization of palaeomagnetic remanence, *Geophys. J. R. astr. Soc.,* **75,** 593–621.

Layer, P., 1986. Archean palaeomagnetism of southern Africa, *PhD thesis,* University of Stanford, Stanford, CA.

McElhinny, M.W., 1966. An improved method for demagnetizing rocks in alternating magnetic fields, *Geophys. J. R. astr. Soc.,* **10,** 369–374.

McElhinny, M.W., Briden, J.C., Jones, D.L. & Brock, A., 1968. Geological and geophysical implications of paleomagnetic results from Africa, *Rev. Geophys.,* **6,** 201–238.

McElhinny, M.W. & Opdyke, N.D., 1964. The palaeomagnetism of the Precambrian dolerites of eastern Southern Rhodesia, an example of geologic correlation by rock magnetism, *J. geophys. Res.,* **69,** 2465–2475.

McFadden, P.L. & Lowes, F.J., 1981. The discrimination of mean directions drawn from Fisher distributions, *Geophys. J. R. astr. Soc.,* **67,** 19–33.

McFadden, P.L. & McElhinny, M.W., 1988. The combined analysis of remagnetization circles and direct observations in palaeomagnetism, *Earth planet. Sci. Lett.,* **87,** 161–172.

McFadden, P.L. & McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism, *Geophys. J. Int.,* **103,** 725–729.

Morgan, G.E., 1976. Palaeomagnetism of a slowly cooled plutonic terrain in western Greenland, *Nature,* **259,** 382–385.

Morgan, G.E., 1985. The palaeomagnetism and cooling history of metamorphic and igneous rocks from the Limpopo Mobile Belt, southern Africa, *Geol. Soc. Am. Bull.,* **96,** 663–675.

Morgan, G.E. & Briden, J.C., 1981. Aspects of Precambrian palaeomagnetism, with new data from the Limpopo Mobile Belt and Kaapvaal craton in southern Africa, *Phys. Earth planet. Inter.,* **24,** 142–168.

Mushayandebvu, M.F., 1991. A palaeomagnetic study of mafic intrusions associated with the Great Dyke, Zimbabwe, *DPhil thesis,* University of Zimbabwe, Harare, Zimbabwe.

Mushayandebvu, M.F., Bates, M.P. & Jones, D.L., 1995a. Anisotropy of magnetic susceptibility results from the Mashonaland dolerite sills and dykes of north-east Zimbabwe, in *Physics and Chemistry of Dykes,* pp. 151–161, eds. Baer, G. & Heimann, A., Balkema, Rotterdam.

Mushayandebvu, M.F., Jones, D.L. & Briden, J.C., 1995b. Palaeomagnetic and geochronological results from Proterozoic mafic intrusions in southern Zimbabwe, in *Physics and Chemistry of Dykes,* pp. 293–303, eds. Baer, G. & Heimann, A., Balkema, Rotterdam.

Pullaiah, G., Irving, E., Buchan, K.L. & Dunlop, D.J., 1975. Magnetization changes caused by burial and uplift, *Earth planet. Sci. Lett.,* **28,** 133–143.

Rollinson, H. & Blenkinsop, T., 1995. The magmatic, metamorphic and tectonic evolution of the Northern Marginal Zone of the Limpopo Belt in Zimbabwe, *J. geol. Soc. Lond.,* **152,** 65–75.

Wilson, J.F., 1991. A craton and its cracks: some of the behaviour of the Zimbabwe Block from the late Archaean to the Mesozoic in response to horizontal movements, and the significance of some of its mafic dyke fracture patterns, *J. Afr. Earth Sci.,* **10,** 483–501.

Wilson, J.F., Jones, D.L. & Kramers, J.D., 1987. Mafic dyke swarms of Zimbabwe, in *Mafic dyke swarms,* pp. 433–444, eds. Halls, H.C & Fahrig, W.F., *Geol. Assoc. Canada, Spec. Paper,* **34.**

Zhang, B. & Halls, H.C., 1995. The origin and age of feldspar clouding in the Matachewan dyke swarm, Canada, *in Physics and Chemistry of Dykes,* pp. 171–176, eds. Baer, G. & Heimann, A., Balkema, Rotterdam.